R Programming Practical Examination

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Part A - Basic Operation A1

```
v <- seq(5:50)
length(v)
```

```
## [1] 46
```

```
sum(v)
```

```
## [1] 1081
```

```
mean(v)
```

```
## [1] 23.5
```

A2

```
w <- rep(c(1,2,3))
rep(c(1,2,3), times = 4)
```

```
##  [1] 1 2 3 1 2 3 1 2 3 1 2 3
```

```
z <- rep(c(1,2,3))
rep(c(1,2,3), each = 4)
```

```
##  [1] 1 1 1 1 2 2 2 2 3 3 3 3
```

A3

```
remainder <- 17 %% 15
int_d <- 17 %/% 15
power <- 2^10
square_rt <- sqrt(144)
pi
```

```
## [1] 3.141593
```

```
remainder
```

```
## [1] 2
```

```
int_d
```

```
## [1] 1
```

```
power
```

```
## [1] 1024
```

```
square_rt
```

```
## [1] 12
```

```r
pi
```

```
## [1] 3.141593
```

A4

```r
a <- c(2, 4, 6, 8)
b <- c(1, 3, 5, 7)
a + b
```

```
## [1]  3  7 11 15
```

```r
a * b
```

```
## [1]  2 12 30 56
```

```r
a > b
```

```
## [1] TRUE TRUE TRUE TRUE
```

```r
sum(a * b)
```

```
## [1] 100
```

A5

```r
paste("Student_1", "Student_2", "Student_3")
```

```
## [1] "Student_1 Student_2 Student_3"
```

```r
typeof(42)
```

```
## [1] "double"
```

```r
typeof(42L)
```

```
## [1] "integer"
```

```r
typeof("42")
```

```
## [1] "character"
```

```r
typeof(41i)
```

```
## [1] "complex"
```

```r
typeof(TRUE)
```

```
## [1] "logical"
```

Part B - Data Frames and Conditional Statement B1

```r
students <- data.frame(
  id = 1:8,
  name = c("Alice", "Bruno", "Chen", "Dana", "Elif", "Farid", "Grace", "Hiro"),
  sex = c("F", "M", "M", "F", "F", "M", "F", "M"),
  section = c("A", "B", "A", "C", "B", "C", "A", "B"),
  midterm = c(78, 85, 92, 67, 74, 88, 95, 59),
  final = c(82, 79, 96, 71, 80, 91, 93, 65),
  attendance = c(95, 88, 100, 72, 90, 85, 98, 60),
  stringsAsFactors = FALSE
)

students
```

```
##   id  name sex section midterm final attendance
## 1  1 Alice   F       A      78    82         95
## 2  2 Bruno   M       B      85    79         88
## 3  3  Chen   M       A      92    96        100
## 4  4  Dana   F       C      67    71         72
## 5  5  Elif   F       B      74    80         90
## 6  6 Farid   M       C      88    91         85
## 7  7 Grace   F       A      95    93         98
## 8  8  Hiro   M       B      59    65         60
```

B2

```
str(students)
```

```
## 'data.frame':    8 obs. of  7 variables:
##  $ id        : int  1 2 3 4 5 6 7 8
##  $ name      : chr  "Alice" "Bruno" "Chen" "Dana" ...
##  $ sex       : chr  "F" "M" "M" "F" ...
##  $ section   : chr  "A" "B" "A" "C" ...
##  $ midterm   : num  78 85 92 67 74 88 95 59
##  $ final     : num  82 79 96 71 80 91 93 65
##  $ attendance: num  95 88 100 72 90 85 98 60
```

```
dim(students)
```

```
## [1] 8 7
```

```
colnames(students)
```

```
## [1] "id"         "name"       "sex"        "section"    "midterm"
## [6] "final"      "attendance"
```

```
summary(students)
```

```
##        id              name               sex             section     
##  Min.   :1.00   Length   :8   Length   :8   Length   :8   Min.   :59.00
##  1st Qu.:2.75   N.unique :8   N.unique :2   N.unique :3   1st Qu.:72.25
##  Median :4.50   N.blank  :0   N.blank  :0   N.blank  :0   Median :81.50
##  Mean   :4.50   Min.nchar:4   Min.nchar:1   Min.nchar:1   Mean   :79.75
##  3rd Qu.:6.25   Max.nchar:5   Max.nchar:1   Max.nchar:1   3rd Qu.:89.00
##  Max.   :8.00                                           Max.   :95.00
##      final         attendance
##  Min.   :65.00   Min.   : 60.00
##  1st Qu.:77.00   1st Qu.: 81.75
##  Median :81.00   Median : 89.00
##  Mean   :82.12   Mean   : 86.00
##  3rd Qu.:91.50   3rd Qu.: 95.75
##  Max.   :96.00   Max.   :100.00
```

B3

```
students$average <- round((students$midterm + students$final) / 2, 1)
students$average
```

```
## [1] 80.0 82.0 94.0 69.0 77.0 89.5 94.0 62.0
```

B4

```
students$passed <- (students$average >= 70) & (students$attendance >= 75)
students$passed
```

```
## [1]  TRUE  TRUE  TRUE FALSE  TRUE  TRUE  TRUE FALSE
```

B5

```
students$standing <- ifelse(students$average >= 90, "Honors",
ifelse(students$average >= 75, "Good", "Needs Support"))

students$standing
```

```
## [1] "Good"          "Good"          "Honors"        "Needs Support"
## [5] "Good"          "Good"          "Honors"        "Needs Support"
```

B6

```
classify <- function(score) {
  if (score >= 85) {
    return("Distinction")
  } else if (score >= 70) {
    return("Merit")
  } else if (score >= 50) {
    return("Pass")
  } else {
    return("Fail")
  }
}
print(sapply(c(90, 73, 55, 40), classify))
```

```
## [1] "Distinction" "Merit"       "Pass"        "Fail"
```

Part C - rbind() and cbind() C1

```
new_students <- data.frame(
  id = c(9, 10),
  name = c("Ines", "Jamal"),
  sex = c("F", "M"),
  section = c("C", "A"),
  midterm = c(81, 70),
  final = c(84, 68),
  attendance = c(92, 80),
  stringsAsFactors = FALSE
)
new_students$average <- round((new_students$midterm + new_students$final) / 2, 1)
new_students$passed <- (new_students$average >= 70) & (new_students$attendance >= 75)
new_students$standing <- ifelse(new_students$average >= 90, "Honors",
ifelse(new_students$average >= 75, "Good", "Needs Support"))

new_students
```

```
##   id  name sex section midterm final attendance average passed      standing
## 1  9  Ines   F       C      81    84         92    82.5   TRUE          Good
## 2 10 Jamal   M       A      70    68         80    69.0  FALSE Needs Support
```

```
new_students$average
```

```
## [1] 82.5 69.0
```

```
new_students$passed
```

```
## [1]  TRUE FALSE
```

```
new_students$standing
```

```
## [1] "Good"          "Needs Support"
```

```
students_all <- rbind(students, new_students)
```

C2

```
quiz <- c(8, 7, 10, 6, 7, 9, 10, 5, 8, 6)
students_all <- cbind(students_all, quiz = quiz)
print(class(students_all))
```

```
## [1] "data.frame"
```

C3

```
a <- cbind(midterm = students_all$midterm, final = students_all$final)
mean_row <- c(mean(students_all$midterm), mean(students_all$final))
a_with_mean <- rbind(a, mean = mean_row)
print(a_with_mean)
```

```
##      midterm final
##         78.0  82.0
##         85.0  79.0
##         92.0  96.0
##         67.0  71.0
##         74.0  80.0
##         88.0  91.0
##         95.0  93.0
##         59.0  65.0
##         81.0  84.0
##         70.0  68.0
## mean    78.9  80.9
```

C4

```
tryCatch(
  rbind(students, bad),
  error = function(e) {
    message("rbind failed: ", conditionMessage(e))
    NULL
  }
)
```

```
## rbind failed: object 'bad' not found
```

```
## NULL
```

Answer(C4): When data frames have mismatched column names or counts, rbind() throws an error because it strictly requires identical column structures to bind rows together. tryCatch() allows the code to handle this exception gracefully without crashing the entire script execution.

Part D - Subsetting D1

```
print(students_all$name)
```

```
##  [1] "Alice" "Bruno" "Chen"  "Dana"  "Elif"  "Farid" "Grace" "Hiro"  "Ines"
## [10] "Jamal"
```

```r
print(students_all[["name"]])
```

```
##  [1] "Alice" "Bruno" "Chen"  "Dana"  "Elif"  "Farid" "Grace" "Hiro"  "Ines"
## [10] "Jamal"
```

```r
print(students_all[, "name"])
```

```
##  [1] "Alice" "Bruno" "Chen"  "Dana"  "Elif"  "Farid" "Grace" "Hiro"  "Ines"
## [10] "Jamal"
```

D2

```r
print(students_all[2:5, c("name", "midterm", "final")])
```

```
##    name midterm final
## 2 Bruno      85    79
## 3  Chen      92    96
## 4  Dana      67    71
## 5  Elif      74    80
```

D3

```r
print(students_all[students_all$attendance > 90, ])
```

```
##   id  name sex section midterm final attendance average passed standing quiz
## 1  1 Alice   F       A      78    82         95    80.0   TRUE     Good    8
## 3  3  Chen   M       A      92    96        100    94.0   TRUE   Honors   10
## 7  7 Grace   F       A      95    93         98    94.0   TRUE   Honors   10
## 9  9  Ines   F       C      81    84         92    82.5   TRUE     Good    8
```